

Sum-Check Toolkit

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Distributed Lab

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Sum-Check-based Protocol for Grand Products

Grand Product Relation

$$\mathcal{R}_{GP} = \left\{ (p \in \mathbb{F}, \mathbf{v} \in \mathbb{F}^m) : p = \prod_{i=0}^m v_i \right\}$$

Assume that m is a power of 2.

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Let $\tilde{v}$ be an MLE of $\mathbf{v}$, by viewing $\mathbf{v}$ as a function mapping $\{0, 1\}^{\log m} \rightarrow \mathbb{F}$.

Main Lemma

Lemma

A scalar p and a vector $\mathbf{v}$ satisfies the relation $\mathcal{R}_{GP}$ if and only if there exists a multilinear polynomial f in $\log m + 1$ variables such that $f(1, \dots, 1, 0) = p$ and $\forall x \in \{0, 1\}^{\log m}$ the following hold:

$$f(0, x) = v(x)$$

$$f(1, x) = f(x, 0) \cdot f(x, 1)$$

Such polynomial f has the following construction:

- $f(1, \dots, 1) = 0$
- For all $\ell \in [\log m]$ and $x \in \{0, 1\}^{\log m - \ell}$:

$$f(1^\ell, 0, x) = \prod_{y \in \{0, 1\}^\ell} v(x, y)$$

Example

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$$v(x_1, x_2) : \quad v(0, 0) = 1, \quad v(0, 1) = 2, \quad v(1, 0) = 3, \quad v(1, 1) = 4.$$

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and:

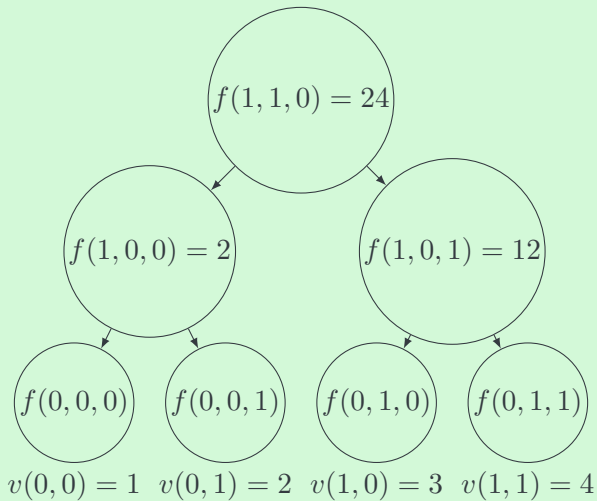
$$f(1, 0, 0) = f(0, 0, 0) \times f(0, 0, 1) = 1 \times 2 = 2,$$

$$f(1, 0, 1) = f(0, 1, 0) \times f(0, 1, 1) = 3 \times 4 = 12,$$

$$f(1, 1, 0) = f(1, 0, 0) \times f(1, 0, 1) = 2 \times 12 = 24 = p,$$

$$f(1, 1, 1) = 0.$$

Example



Where Is Sum-Check?



Zero-Check

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One can use the sum-check protocol to prove the evaluation of g that is referred to a MLE of $f(1, x) - f(x, 0) \cdot f(x, 1)$:

$$g(t) = \sum_{x \in \{0, 1\}^{\log m}} \tilde{\text{eq}}(t, x) \cdot (f(1, x) - f(x, 0) \cdot f(x, 1))$$

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By the Schwartz–Zippel lemma, for random $\tau \in \mathbb{F}^{\log m}$, $g(\tau) = 0$ if and only if $g = 0$, except for a soundness error of $\frac{\log m}{|\mathbb{F}|}$.

Thus, to prove the existence of f and hence the grand product relationship, it suffices to prove, for some verifier selected random $\tau, \gamma \in \mathbb{F}^\ell$, that:

$$0 = \sum_{x \in \{0,1\}^{\log m}} \tilde{eq}(x, \tau) \cdot (f(1, x) - f(x, 0) \cdot f(x, 1)) \quad (1)$$

$$f(0, \gamma) = \tilde{v}(\gamma) \quad (2)$$

$$f(1, \dots, 1, 0) = p \quad (3)$$

Algorithm 1: Sum-Check-based Protocol for Grand Products

- 1 $\mathcal{P}$: Compute polynomials $v \in \mathbb{F}^{\log m}[x]$, $f \in \mathbb{F}^{\log m+1}[x]$ such that

$$p = \prod_{x \in \{0,1\}^{\log m}} v(x) \quad \text{and} \quad f, v \text{ satisfy (1), (2), (3).}$$

- 2 $\mathcal{P}$: $C_f \leftarrow \text{Commit}(f)$; $C_v \leftarrow \text{Commit}(v)$; send C_f, C_v to $\mathcal{V}$.
 3 $\mathcal{V}$: Choose random $\tau, \gamma \in \mathbb{F}^{\log m}$ and send them to $\mathcal{P}$.
 4 $\mathcal{P}$: Compute $g(x) = \tilde{\text{eq}}(x, \tau) (f(1, x) - f(x, 0) f(x, 1))$.
 5 $\mathcal{P} \ \& \ \mathcal{V}$: Run $\text{SumCheckProtocol}(0, g, C_f)$
 6 $\mathcal{V}$: $a \leftarrow \text{Query}(C_f, (0, \gamma))$, $v(\gamma) \leftarrow \text{Query}(C_v, \gamma)$.
 7 **if** $a \neq v(\gamma)$ **then**
 8 | $\mathcal{V}$ **rejects**.
 9 **end**
 10 $\mathcal{V}$: $r \leftarrow \text{Query}(C_f, (1, \dots, 1, 0))$.
 11 **if** $r \neq p$ **then**
 12 | $\mathcal{V}$ **rejects**.
 13 **end**
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Randomized Permutation Check

The goal is to verify, with high probability, whether two sequences of tuples are permutations of each other, without performing a full sort or pairwise comparison.

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$$A = \{(1, 2, 3), (4, 0, 6)\}, \quad B = \{(4, 0, 6), (1, 2, 3)\}.$$

The naive approach would be to sort both sequences and then compare them.

Reed-Solomon Fingerprinting

Definition (Reed-Solomon Fingerprinting)

Let $\mathbf{a} \in \mathbb{F}^n$, then for a random $\gamma \in \mathbb{F}$, the Reed-Solomon fingerprinting of $\mathbf{a}$ is defined as:

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$h_{\gamma}(\mathbf{a})$ uniquely identifies the sequence $\mathbf{a}$ with high probability, i.e., let $\mathbf{b} \in F^n$ and $\mathbf{a} \neq \mathbf{b}$, then, according to the Schwartz-Zippel lemma:

$$\Pr[h_{\gamma}(\mathbf{a}) = h_{\gamma}(\mathbf{b})] \leq \frac{n}{|\mathbb{F}|}.$$

Reed-Solomon Fingerprinting

Example

Consider all operations in $\mathbb{F}_7$, and set $n = 3$, $\gamma = 3$. Let

$$\mathbf{a} = (1, 2, 3), \quad \mathbf{b} = (4, 0, 6).$$

Then

$$h_\gamma(1, 2, 3) = 1 \cdot 3^0 + 2 \cdot 3^1 + 3 \cdot 3^2 = 1 + 6 + 6 = 13 \equiv 6,$$

$$h_\gamma(4, 0, 6) = 4 \cdot 1 + 0 \cdot 3 + 6 \cdot 2 = 4 + 0 + 12 = 16 \equiv 2.$$

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Let A and B be two multisets of tuples in $\mathbb{F}^n$. Define

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Then comparing $\mathcal{H}_{\tau,\gamma}(A)$ and $\mathcal{H}_{\tau,\gamma}(B)$ yields a randomized test for whether A and B are permutations of one another. Concretely:

- (Completeness) If $A = B$ (as multisets), then

$$\mathcal{H}_{\tau,\gamma}(A) = \mathcal{H}_{\tau,\gamma}(B)$$

with probability 1 over uniform $\tau, \gamma \in \mathbb{F}$.

- (Soundness) If $A \neq B$, then

$$\Pr[\mathcal{H}_{\tau,\gamma}(A) = \mathcal{H}_{\tau,\gamma}(B)] \leq \frac{\max(|A|, |B|)}{|\mathbb{F}|}.$$

Example

Consider all operations in $\mathbb{F}_7$, set $n = 3$, $\tau = 5$, and $\gamma = 3$.
Let $A = \{(1, 2, 3), (4, 0, 6)\}$, $B = \{(4, 0, 6), (1, 2, 3)\}$.

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First compute the Reed–Solomon fingerprints modulo 7:

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Now form the shifted products:

$$\mathcal{H}_{\tau, \gamma}(A) = (6 - 5)(2 - 5) = 1 \cdot (-3) \equiv 4,$$

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Now consider a non-permutation $B' = \{(1, 2, 3), (2, 1, 3)\}$.

$$h_\gamma(2, 1, 3) = 2 \cdot 1 + 1 \cdot 3 + 3 \cdot 2 = 2 + 3 + 6 = 11 \equiv 4.$$

$$\mathcal{H}_{\tau, \gamma}(B') = (6 - 5)(4 - 5) = 1 \cdot (-1) \equiv 6 \neq 4,$$

so the test *rejects* A vs. B' .

Offline Memory Checking

Motivation

Consider Alice, who stores two values on Bob's dedicated server at addresses 0 and 1. Initially, Bob's memory contains

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However, Bob can cheat on the very last step by returning a wrong value:

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Without keeping an auditable record of all reads, Alice cannot later prove that all replies came from correct memory contents.

Memory Model

Each **memory cell** can be described as a tuple (addr, val, counter).

- addr is the address of the memory cell;
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The protocol utilizes four sets of tuples:

- **init** – contains the initial memory state;
- **write** – contains memory cels that represent write operations;
- **read** – contains memory cels that represent read operations;
- **final** – contains the final memory state, where all counters are set to the last value.

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- **Read:**
 - Query untrusted memory at $\text{addr} \rightarrow (\text{val}, \text{counter})$;
 - Append $(\text{addr}, \text{val}, \text{counter})$ to **reads**;
 - Append $(\text{addr}, \text{val}, \text{counter}+1)$ to **writes**.

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- **Read:**
 - Query untrusted memory at `addr` $\rightarrow$ `(val,counter)`;
 - Append `(addr,val,counter)` to **reads**;
 - Append `(addr,val,counter+1)` to **writes**.
- **Write:**
 - Query untrusted memory at `addr` $\rightarrow$ `(val,counter)`;
 - Append `(addr,val,counter)` to **reads**;
 - Append `(addr,newval,counter+1)` to **writes**.

Consistency Check

After all reads and writes are done, the **final** set is populated with the final memory state.

Lemma

One can check the consistency of the memory operations by verifying that:

$$\mathbf{read} \cup \mathbf{final} = \mathbf{write} \cup \mathbf{init}.$$

Equivalently, via randomized permutation check:

$$\mathcal{H}_{\tau,\gamma}(\mathbf{read}) \cdot \mathcal{H}_{\tau,\gamma}(\mathbf{final}) = \mathcal{H}_{\tau,\gamma}(\mathbf{write}) \cdot \mathcal{H}_{\tau,\gamma}(\mathbf{init}).$$

Example

Suppose $N^{1/c} = 4$, $m = 3$, and the initial memory is:

$$\mathbf{init} = \{(0, 2, 0), (1, 5, 0), (2, 7, 0), (3, 9, 0)\},$$

while $\mathbf{read} = \emptyset$ and $\mathbf{write} = \emptyset$.

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step	operation	$\Delta\text{read}_{\text{step}}$	$\Delta\text{write}_{\text{step}}$
1	$\text{read}(1) \rightarrow (5, 0)$	$(1, 5, 0)$	-
2	$\text{write}((1, 6))$	-	$(1, 6, 1)$
3	$\text{read}(2) \rightarrow (7, 0)$	$(2, 7, 0)$	-
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4	write((2,7))	-	(2, 7, 1)

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One can clearly see that $\mathbf{read} \cup \mathbf{final} = \mathbf{write} \cup \mathbf{init}$.

$$\begin{aligned} & \{(1, 5, 0), (2, 7, 0)\} \cup \{(0, 2, 0), (1, 6, 1), (2, 7, 1), (3, 9, 0)\} = \\ & = \{(1, 6, 1), (2, 7, 1)\} \cup \{(0, 2, 0), (1, 5, 0), (2, 7, 0), (3, 9, 0)\} \end{aligned}$$

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The verifier checks $\mathbf{read} \cup \mathbf{final} = \mathbf{write} \cup \mathbf{init}$:

$$\begin{aligned} & \{(1, a, 0)\} \cup \{(0, 2, 0), (1, 6, 1), (2, 7, 0), (3, 9, 0)\} \neq \\ & \neq \{(1, 6, 1)\} \cup \{(0, 2, 0), (1, 5, 0), (2, 7, 0), (3, 9, 0)\} \end{aligned}$$

and *rejects*.

Questions?